Mathematical overview

For these polygonal shapes, the count of grid cells wholly gray is a lower area bound and the count containing positive gray area is an upper bound, with one cell as area unit. Boundary-only contact adds no cover cell. Translating a grid at fixed cell size is not nested refinement: for the fixed area-3 L polygon, the aligned bounds are [3,3], a half-unit horizontal shift gives [1,5], and a half-unit shift in both directions gives [0,8]. Moving from the aligned grid worsens both; reversing that move improves them. No monotonic improvement theorem holds for arbitrary translations.

Two guaranteed area ranges [L1,U1] and [L2,U2] for the same unchanged shape combine to [max(L1,L2),min(U1,U2)]. A reversed interval means inconsistent reports. This is intersection of deterministic possibilities, with no averaging or probability model. The cards are guarantees, not necessarily exact whole/partial cell counts from one specified grid.

Within the printed 2-by-2 board, the three-cell L mask permits connected hole-free polygons with arbitrarily small positive area or area arbitrarily close to 3, while every marked cell remains genuinely partial. A thick L corridor of width t, $0<t<1$, has area $4t-t^2$ and occupies exactly those three cells. Equal area and the same mask also permit arbitrarily large perimeter: replace a straight boundary segment with balanced triangular waves whose signed area changes cancel. Each construction is finite. Finite printing resolution limits physical examples; it does not make the ideal polygonal statement false.

Grades 2-5 can compare grids and combine guarantees; grades 4-5 can distinguish area from boundary length and explain equal-area modifications. Measurements and drawings supply experiments. Cell inclusion, intersections, and exact polygon calculations establish claims; a few jagged drawings alone do not prove unbounded perimeter.

Readiness, kits, launch, and return visits

Unpiloted; transparent-grid handling, tracing and exact printing have not been physically tested. Page 1 needs identifying whole/partial cells and counting through 8. Page 2 needs interpreting lower and upper guarantees, with adult reading as needed. Page 3 needs distinguishing area from perimeter and comparing small fractions through drawings; its deeper argument suits ready grades 4-5. K-1 can join concrete counting on page 1, but a separate independent version is not forced.

Per pair: the packet, a transparent sheet or tracing paper, ruler, pencil and colored pencil. Trace a grid with 56-point (19.8 mm) sides for the page-1 diagrams; 19.8 mm is the same in both axes. For larger manipulation, draw a 30 mm grid and an L from three 30 mm squares on scrap, then slide the overlay; this is a faithful enlargement, not an additional printed asset. Keep the shape stationary. Page 2 now supplies two 6-by-4 witness grids with 27-point (9.5 mm) unit-square sides, one for each possible pair (the first and third of the numbered pairs 1-3), each captioned "Pair: ____" for the child to write its number, so the page does not say which pair is impossible; use these for checkable shape drawings. Page-3 cells have 80-point sides (28.2 mm). Twelve paper tiles can support cover counting but are optional; shading is sufficient.

For eleven children at K–1 / 2–3 / 4–5: five pair kits, hence five overlays, five rulers, ten pencils, five colored pencils, and optionally sixty paper square tiles. The third older child checks cell status then rotates roles. Three anchored adults. K-1 may return to the base cell-cover pages while older groups use these companions.

Common launch, 3-4 minutes: let children lay a grid over a shape. Use the page-2 partial-square visual to show why its card is 0 to 1, without claiming the triangle's exact area from the card. Point out edge-only contact on one square. Demonstrate sliding the overlay while the shape stays fixed; do not announce whether it improves the estimate. For the mask page, distinguish a marked cell from a gray-filled cell: partial means both colors have positive area.

First route: 15-25 minutes on shifted grids, then 10-20 on cards. Reserve 20-40 minutes on a later visit for mask/perimeter designs. The third investigation offers continued depth rather than a required finish. Save both shape and grid origin in a record; a count without its placement cannot be checked. Exact-fit cutting is not required.

Solutions, hints, and continuations

1. The fixed polygon is the union of [0,2]x[0,1] and [1,2]x[1,2], with area 3. Aligned: 3 inside, 3 covered. Horizontal half shift: the middle bottom square is whole, two bottom edge squares and two top squares are partial, hence [1,5]. Both half shifts: 0 whole and 8 covered. The aligned grid gives an exact certificate; translations can make it worse. Optional hint: color cells that only touch separately from cells that contain gray. Extension: choose a shape with sloping edges and compare several translations; ask whether two translated certificates can help different ends of a range.

2. Cards 2-8 and 4-9 combine to 4-8. Cards 3-5 and 6-7 are inconsistent: the area would be simultaneously at most 5 and at least 6. Cards 1-6 and 4-4 combine to exactly 4. For possible pairs, a 2-by-2 unit square of area 4 works for both; a 2-by-3 rectangle of area 6 also works for the first. These are witnesses, not unique shapes. Endpoints are allowed. Optional hint: ask which areas one report rules out, while keeping both reports about the same object. Extension: introduce a third card or ask children to invent two cards, each of which contributes one endpoint of the combined range.

3. Use the polygon with vertices (0,0),(2,0),(2,2),(2-t,2),(2-t,t),(0,t). Its cell areas, bottom-left, bottom-right, top-right, are t,$2t-t^2$,t, all strictly between 0 and 1. At $t=1/4$ total is $15/16<1$. At $t=3/4$ total is $39/16>2$. Both are connected and have no holes. Their perimeters happen to both be 8; area and perimeter need separate reasoning.

For an equal-area longer-boundary example, keep $t=1/4$ and replace the horizontal top edge over $0<=x<=1$ in the bottom-left cell. For N waves use vertices with successive x spacings $1/(4N)$, heights t,$t+1/8$,t,$t-1/8$,t repeated. The positive and negative triangles cancel, so area remains 15/16; the height remains between 1/8 and 3/8, hence the same cell mask and connectivity persist. The changed edge has length $\sqrt{1+N^2/4}$, so full perimeter is $7+\sqrt{1+N^2/4}$, exceeding any chosen bound for sufficiently large finite N. Children can argue that each up/down tooth contributes a fixed vertical amount, without calculating square roots. Optional hint: trade equal small triangles above and below one straight edge. Extension: set a minimum tooth size or number of vertices and investigate the new finite design limit. Holes and disconnected gray islands are forbidden in the printed task.

New versus base and check evidence

Base Week 48 uses nested grid refinement and rectangular masks with different areas. Page 1 translates the grid at unchanged scale; page 2 combines independent guarantees rather than refining cells; page 3 changes to a connected L mask and adds a new equal-area/unbounded-perimeter direction. The first area contrast on page 3 is an entry into this fresh perimeter investigation, not counted as another investigation.

Original problems extending the base finite-cover mathematics. Math Circle by the Bay, preface printed pp. viii-x (PDF pp. 9-11), supports flexible return visits; it is not the source of these exact tasks. student/verify.py uses exact rational rectangle intersections for all shifted cover counts, checks card intersections, computes both explicit witness areas and every cell status, and checks exact area cancellation plus boundary lengths for 1,4,20 triangular-wave examples. Unboundedness follows from the formula, not those three samples. No classroom or physical test is claimed.